Assignment 2: Arrays and Linked Lists in LC-3

CSCI260: Computer Architecture, Fall2025 **Topic:** Data Structures at the ISA Level **Due: 11/17, 23:50**

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1. Objective

By the end of this assignment, you should be able to:

- Explain what **arrays** and **linked lists** are in your own words.
- Describe their **features** and **typical operations/behaviors**.
- Map high-level data structure concepts onto **LC-3 ISA instructions**.
- Write **well-organized, well-documented LC-3 code snippets** that implement basic array and linked list behaviors.

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2. Assignment Description

This assignment has two main parts.

Part A – Concept Reference (High-Level View)

1. **Look up** the definition of **Array** and **Linked List** in any language you are comfortable with (e.g., English, your native language).

2. For each data structure, create a **one-page reference** (so: one page for Array, one page for Linked List). Each page must include:

    - **Definition**
    - **Key Features**
        - e.g., memory layout, access pattern, typical use cases
    - **Operations / Behaviors**
        - e.g., access element, insert, delete, traverse, search
    - **Optional but encouraged:** graphics, diagrams, memory layouts, arrows showing links, etc.

Your goal is to show that you **understand** arrays and linked lists at a high level and can summarize them clearly.

Part B – LC-3 Translation (ISA Level)

Based on your references from Part A:

1. **Translate the definition and behavior** of the **array** and the **linked list** into **LC-3 code snippets** (ISA-level instructions).

2. For each data structure, write LC-3 code that demonstrates at least:

    - How the structure is **represented in memory**.
    - How one or more **basic operations** work. For example:
        - **Array**:
            - Access element at index i
            - Update an element
            - Traverse all elements
        - **Linked list**:
            - Traverse the list
            - Insert at head (or another simple operation)
            - Search for a value

3. In your LC-3 programs:

    - Use **labels** wisely to organize your code (e.g., labels for loops, functions, data sections).
    - **Clearly separate** code for arrays and linked lists.
    - Add **comments/documentation** that:
        - Explain what each snippet does
        - Describe how to “use” it (inputs, outputs, assumptions)
        - Connect the LC-3 instructions back to the behaviors described in your reference pages

The focus is on showing how a high-level data structure can be implemented using **low-level instructions and memory addresses**.

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3. Deliverables

You will submit the following:

1. **Reference Pages (PDF or similar)**

    - **1 page for Array**
    - **1 page for Linked List**
    - Each page must include:
        - Definition
        - Features
        - Operations/Behaviors
        - Optional graphics/diagrams (strongly encouraged)

2. **LC-3 Code Snippets + Documentation**
    - LC-3 **source files** (e.g., `.asm`) for:
        - Array-related code
        - Linked-list-related code
    - Short **documentation** (can be in comments inside the code or in a separate text/PDF) explaining:
        - How to use each code snippet
        - What each label/function does
        - How the code corresponds to the behaviors you described in the reference pages

Recommended submission format (one ZIP file):

- `array_reference.pdf`
- `linked_list_reference.pdf`
- `array.asm`
- `linked_list.asm`
- (Optional) `readme.txt` or `documentation.pdf`

Upload this ZIP file to the course page under this assignment.

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4. Grading Criteria

Your work will be evaluated using the following criteria:

4.1 Conceptual Understanding (30%)

- Definitions of array and linked list are accurate and clear.
- Features and behaviors are correctly described.
- Differences between array and linked list are apparent from your references.

4.2 Completeness & Organization of Reference Pages (20%)

- Each data structure has a **separate, one-page** reference.
- All required sections (Definition, Features, Operations/Behaviors) are present.
- Information is well-organized and easy to read.
- Use of diagrams/graphics to illustrate memory layout and/or behavior (not required, but rewarded).

4.3 LC-3 Implementation Correctness (25%)

- LC-3 code correctly represents arrays and linked lists in memory.
- Basic operations behave as described (e.g., traversal, access, simple insert/search).
- Code assembles and runs logically (minor syntax issues may be tolerated, but semantic correctness matters).

4.4 Code Organization & Use of ISA Constructs (15%)

- Effective and meaningful use of **labels** for structure and readability.
- Clear separation of code sections (data, code, different operations).
- Good use of LC-3 instructions and addressing modes to implement behaviors.

4.5 Documentation & Style (10%)

- Comments and/or separate documentation clearly explain how to use the snippets.
- Connection between high-level description and ISA-level code is clear.
- Overall presentation is tidy and professional.

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5. Notes

- You may write the reference pages in any **natural language** you are comfortable with.
- LC-3 code and labels should be written clearly so that the instructor and TAs can understand them.
- You are encouraged to collaborate in discussing concepts, but all submitted code and text must be **your own work**.